

```

out = 4'b0;
else
out = {out[2:0], feedback};
end
endmodule

```

Testing and simulation

To simulate the LFSR design, install ModelSim V10.4a on a Windows PC and follow the steps given below:

1. Start ModelSim from the desk-

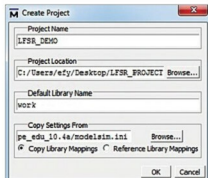


Fig. 2: Create Project window

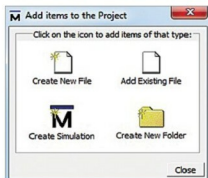


Fig. 3: Add Items to the Project window

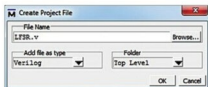


Fig. 4: Create Project File

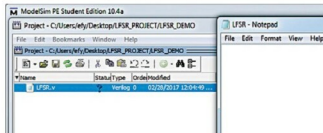


Fig. 5: Workspace window

top; you will see ModelSim 10.4a dialogue window

2. Create a project by clicking 'JumpStart' in the welcome screen

3. In 'Create Project' window (Fig. 2), select a suitable name for your project. Set 'Project Location' and leave the rest as default, then click 'Ok.' 'Add items to the Project' window pops up (Fig. 3).

4. In 'Add items to the Project' window, select 'Create New File' option

5. In 'Create Project File' window, give an appropriate file name

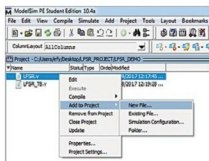


Fig. 6: Adding a new file

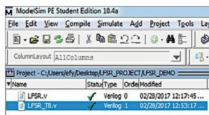


Fig. 7: Compilation window

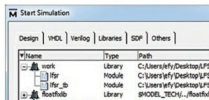


Fig. 8: Library tab

(say, lfsr.v) for the file you want to add, and choose Verilog under 'Add file as type' and 'Top Level' under 'Folder' (Fig. 4)

6. In the workspace section of the main window (Fig. 5), double-click the file you have just created (lfsr.v in our case)

7. Type your Verilog code in the new window. The main goal here is to write a self-test bench that will generate clock automatically for the simulation output window. Save your code from 'File' menu

8. Now, add the new file to lfsr.v by right-clicking it. Select Add to Project→New File as shown in Fig. 6

Give the file name as lfsr_tb.v. This is the test case where you can test and understand the working of a linear-feedback shift register. Details of this LFSR can be understood from the relevant source file.

Compiling/debugging project files.

1. Select Compile→Compile All
2. Compilation result is shown on the main window. A green tick against each file name indicates that there are no errors in the project (Fig. 7)

Simulating 4-bit LFSR.

1. Select Simulate→Start Simulation. Click 'Library' menu from the main window, followed by plus (+) sign next to the work library. You should see the name that you have just compiled in Fig. 7

2. In work library, select lfsr_tb and click 'Ok' (Fig. 8). This opens sim-Default window as shown in Fig. 9

3. Go to Add→To Wave→All items in region

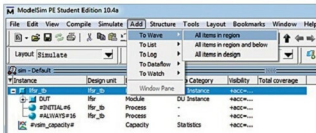


Fig. 9: Add wave to the project